

# Multicopter Physical Specifications Questionnaire

For each airframe in the \_Tuned.af files, please provide the following physical specifications so we can accurately model the dynamics in simulation.

## Required Parameters

| Parameter               | Unit              | Description                                  | Example |
|-------------------------|-------------------|----------------------------------------------|---------|
| PHYS_MASS_KG            | kg                | Total AUW (All-Up Weight) including battery  | 0.91    |
| PHYS_FRAME_MM           | mm                | Diagonal motor-to-motor distance             | 220     |
| PHYS_MOTOR_KV           | RPM/V             | Motor KV rating                              | 1000    |
| PHYS_PROP_DIAMETER_INCH | inch              | Propeller diameter                           | 11      |
| PHYS_PROP_PITCH_INCH    | inch              | Propeller pitch                              | 3.7     |
| PHYS_PROP_BLADES        | count             | Number of blades per prop (2, 3, 4)          | 2       |
| PHYS_CELL_COUNT         | S                 | Battery cell count (1S, 2S, 3S, 4S, 6S)      | 4       |
| PHYS_BATTERY_MAH        | mAh               | Battery capacity                             | 1500    |
| PHYS_ARM_LEN_M          | m                 | Center-to-motor distance                     | 0.11    |
| PHYS_INERTIA_ROLL_PITCH | kg·m <sup>2</sup> | Roll/Pitch moment of inertia                 | 0.002   |
| PHYS_INERTIA_YAW        | kg·m <sup>2</sup> | Yaw moment of inertia                        | 0.005   |
| PHYS_MAX_THRUST_N       | N                 | Max thrust per motor (measured or estimated) | 3.5     |

## Optional (Fixed Wing Only)

| Parameter            | Unit | Description               |
|----------------------|------|---------------------------|
| PHYS_CRUISE_SPEED_MS | m/s  | Cruise airspeed           |
| PHYS_MIN_SPEED_MS    | m/s  | Minimum airspeed (stall)  |
| PHYS_MAX_SPEED_MS    | m/s  | Maximum airspeed          |
| PHYS_LIFT_COEFF      | -    | Wing lift coefficient     |
| PHYS_DRAG_COEFF      | -    | Parasite drag coefficient |

## Airframes Needing Specs

### Ecks Series (all 2208 1000kv, 11x3.7" 2-blade, 4S)

- ☐ **EcksTuned.af** — 910g AUW, 220mm frame
- ☐ **Ecks\_220mm\_Tuned.af** — same as above
- ☐ **Ecks\_800g\_Moderate.af** — 800g AUW, 220mm frame
  
- ☐ **Ecks\_800g\_Sport.af** — 800g AUW, 220mm frame
- ☐ **Ecks\_1kg\_Moderate.af** — 1kg AUW, BR2205 2300kv + Cyclone 5x3x3
- ☐ **Ecks\_1kg\_Sport.af** — 1kg AUW, BR2205 2300kv + Cyclone 5x3x3
- ☐ **EcksQuatQ.af** — quaternion retune

## Ken's Models

- ☐ **Ken\_s\_Alpha\_Test\_Tuned.af** — LOW\_VOLT\_THRES=13.8V (4S?), 450mm frame?
- ☐ **Ken\_s\_450\_1165\_Tuned.af** — 450-1165 frame, what motors/props?
- ☐ **Ken\_s\_LadyBug\_Tuned.af** — WLToys brushed 1S, what frame/motors?

## Other

- ☐ **DevEBox\_Tuned.af** — what hardware?
  - ☐ **S500\_1137\_Tuned.af** — S500 frame, what motors/props?
  - ☐ **150mm\_Brushed\_Tuned.af** — 150mm brushed, 1S/2S?
  - ☐ **Rok\_s\_Quad\_Tuned.af** — AF\_TYPE=QUAD (not QUAD\_X), LOW\_VOLT\_THRES=10V (3S?), BATTERY\_CAPACITY=50 (5000mAh?), ESC=FAST\_PWM, what frame/motors/props?
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## How to Fill

1. **Mass:** Weigh with battery installed (AUW)
  2. **Frame:** Measure motor-to-motor diagonal
  3. **Motor:** Check label or invoice for KV
  4. **Prop:** Check prop markings (e.g., "11x3.7" = 11" dia, 3.7" pitch)
  5. **Battery:** Check label for S-count and mAh
  6. **Arm length:** Half of frame diagonal (or measure center to motor)
  7. **Inertia:** If unknown, use approximation:
    - $I_{roll\_pitch} \approx mass \times (arm\_len)^2 / 12$
    - $I_{yaw} \approx 2.5 \times I_{roll\_pitch}$
  8. **Max thrust:** If unknown, estimate from motor+prop data or use  $(mass \times 9.81) / (hover\_throttle \times num\_motors)$
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## Output Format

Please provide as:

EcksTuned.af:

```
PHYS_MASS_KG = 0.91
PHYS_FRAME_MM = 220
PHYS_MOTOR_KV = 1000
PHYS_PROP_DIAMETER_INCH = 11
PHYS_PROP_PITCH_INCH = 3.7
PHYS_PROP_BLADES = 2
PHYS_CELL_COUNT = 4
PHYS_BATTERY_MAH = 1500
PHYS_ARM_LEN_M = 0.11
PHYS_INERTIA_ROLL_PITCH = 0.002
PHYS_INERTIA_YAW = 0.005
PHYS_MAX_THRUST_N = 3.5
```

Save responses to: uavx-python/src/airframes/SPECS\_RESPONSES.md